

The evolution of market share among the U.S. options market platforms

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Highlights

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Sample period includes the opening of two additional options exchanges.

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Market share is positively related to execution quality.

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Market share is positively related to quote competition variables.

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Market share is negatively related to execution costs.

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Exchanges using the maker–taker pricing model account for lower trading volume.

Abstract

The U.S. equity options market has shown rapid growth, characterized by increased trading volume and new entrants in recent years. As competition in the options market has intensified, each trading venue has attempted to differentiate itself with a unique value proposition aimed at gaining market share. Using a sample that covers key events including the introduction of two new trading venues, we analyze the competition for trading volume, investigate the characteristics of trades and execution costs on the exchanges and examine the evolution of market share in the options industry. We find that exchange market share is positively related to execution quality and quote competition indicators and negatively related to execution costs, while

exchanges that use the maker–taker model as their primary pricing model account for, on average, a lower market share than exchanges that use alternative pricing models. Given that exchanges increasingly compete using innovative methods involving market structure such as different order types, fee structures, pricing methods, and priority rules for matching orders, our finding that higher market quality in the form of higher execution quality, lower execution costs, and competitive quotes translates into higher market share is especially noteworthy.

Introduction

In recent years, the U.S. equity options market has shown rapid growth characterized by increased trading volume and new entrants. As of 2011 there were ten options exchanges operating in the United States: NYSE Arca, NYSE Amex, The Boston Options Exchange (BOX), The Chicago Board Options Exchange (CBOE), The International Securities Exchange (ISE), NASDAQ OMX PHLX (PHLX), NASDAQ Options Market (NOM), BATS Options Exchange (BATS), CBOE Options Exchange (C2), and NASDAQ OMX BX. Intense competition, leading to a fast changing market environment, has given rise to a need for research exploring the competition and market structure of modern exchanges (Cantillon & Yin, 2011). We address this need by studying the competition for trading volume, investigating the characteristics of trades and execution costs on the exchanges and examining the evolution of market share in the U.S. options industry.

Competing for a share of the market, options exchanges attempt to differentiate their services through variations in market structure and trading platforms. While some exchanges combine an electronic trading system with an open-outcry trading floor in a hybrid model, others use fully electronic platforms. As competition among the exchanges has evolved, the pricing model has also become an important dimension on which exchanges try to differentiate themselves (see, for example, Birkner, 2010, Cardella et al., 2013, Chapman, 2010, DeCovny, 2008, Hintze, 2009). Some exchanges use payment for order flow to attract trading interest and others have introduced a maker–taker model that is typical of order-driven equity markets in an attempt to carve a niche for themselves in this very competitive landscape. In addition, a taker–maker model, where the incentives for liquidity provision

and consumption are reversed, was introduced as a further means of differentiation by BOX (B) and the recent addition of NASDAQ OMX BX. Table 1, which provides a summary of the characteristics of the U.S. options exchanges during our sample period of January 2008 to December 2010, shows two main types of pricing models: (1) a maker–taker model with price–time priority for order allocation and (2) a payment for order flow driven pro-rata model with order allocation based on customer priority. A pro-rata model is used by NYSE Amex (A), CBOE (C), ISE (I), NASDAQ OMX PHLX (X), while NYSE Arca (N), NASDAQ Options Market (Q), C2 (W), and BATS (Z) use the maker–taker model.

In the payment for order flow driven pro-rata model, market-makers provide continuous two-sided quotes and customer orders are given priority in order execution. Order allocation to liquidity providers is based on the size of their quotes. The market maker's willingness to quote in larger sizes is a relative advantage of the exchanges that use the pro rata model. In the maker–taker model with price–time priority for order allocation, order execution is based on time priority at a given price level, with liquidity takers paying a fee and liquidity providers receiving rebates. This model is common in electronic trading platforms. Battalio, Shkilko, and Van Ness (2011) compare trading costs for maker–taker pricing and payment for order flow exchanges in the U.S. options market. They find that trading costs under the payment for order flow model are lower than those under the maker–taker model. Anand, McCormick, and Serban (2013) compare the performance of the maker–taker model with the traditional model used in option markets. They show that although the maker–taker model covers a small, high-volume subset of listed options it provides better execution quality and leads to improved price quotes. They also show that the traditional model provides better coverage of less liquid securities, more depth and quotes consistently at the NBBO, and may lead to higher realized spreads on options traded under this model.

As intense competition is leading to a fast changing option market environment, it is becoming increasingly important for market participants to understand the market structure and quality in different options trading venues (see, for example, Aite Group, 2008, Babel, 2011a, Babel, 2011b, Beals, 2012, Reuters, 2010). Several studies examine the impact of competition on market quality, especially trading costs, on options exchanges (Battalio et al.,

2004, de Fontnouvelle et al., 2003, de Fontnouvelle et al., 2005, Hansch and Hatheway, 2001, Mayhew, 2002). Hansch and Hatheway (2001) examine market quality in the option exchanges, demonstrating that market quality has changed significantly as multiple exchange listings have become the norm, and indicate that execution quality measures differ across exchanges. Mayhew (2002) investigates the effects of competition and market structure on the bid-ask spreads of stock options traded on the Chicago Board of Options Exchange (CBOE) from 1986 to 1997. He finds that options listed on multiple exchanges have narrower spreads than those listed on a single exchange, but the difference diminishes as option volume increases. He also shows that when there is a delisting of an option from the competitive exchange, the spread widens. Battalio et al. (2004) study the U.S. options market for efficiency of transactions and execution quality. They document significant changes in the market between June 2000 and January 2002, after the beginning of intense competition for order flow during August 1999, the introduction of payment for order flow shortly afterwards, and the launch of the International Securities Exchange in May 2000. Examining option quotes, they report evidence of improved market quality during the time separating their sample periods. de Fontnouvelle et al. (2003), de Fontnouvelle et al. (2005) focus on execution costs during a period when the U.S. options exchanges began to compete directly for order flow in many options classes that had previously been exclusively listed. They show that effective and quoted bid-ask spreads decreased significantly as a result of the competition that began with multiple listings in August 1999 and that spreads generally maintained their lower levels one year later.

The aforementioned studies contribute to the literature by pinpointing the start of heightened competition among the options exchanges and by showing some of the benefits of competition for investors in the form of improved market quality. In doing so, they typically use an event study methodology taking advantage of an event that initiates or enhances competition among the exchanges and report changes in execution costs, most commonly the spreads, as a result of the event. Although these studies establish the connection between the degree of competition and market quality, they do not attempt to explain how industry competition plays out, eventually leading to a separation of winners and losers. The questions of how option markets attract order flow

and market share has evolved among the competing options exchanges have received relatively little attention in the existing literature. In fact, Cantillon and Yin (2011) point out the need for a research agenda related to the competition and market structure of financial exchanges, and argue that a combination of industrial organization and finance research can address this need.

Our study addresses this need in the context of the U.S. option market and explores the competition among the option exchanges during a period that covers important milestones, including the implementation of the Penny Pilot Project (PPP) and the launch of the NASDAQ Options Market and the BATS Options Exchange. Specifically, we study the evolution of market share using a panel data set aggregated at the monthly level to establish the link between market share and execution costs, execution quality, and quote competition variables for exchanges with different market structures. We focus on market share as the dependent variable and investigate two main hypotheses. First, we use a regression framework to test the hypothesis that market share is positively correlated with trade execution quality variables and that it has a negative relationship with execution costs. We find that a positive relationship between market share and trade execution quality persists during our three-year sample period and that winners in the competition for market share have lower execution costs. Our second hypothesis is that there is a positive relationship between market share and competitive quotes. We show that providing competitive quotes increases market share of an exchange. Overall, our results indicate that market share is higher when a market provides lower execution costs, has better trade execution quality compared to the NBBO at the time of a trade, and is the only market with the best quotes at the time of a trade. Additional analysis using more granular intra-day data for the NASDAQ Options Market and the BATS Options Exchange and examining how new venues establish a foothold in the market in their beginning days confirms our findings. We find that new venues competing to establish market share have a higher likelihood of attracting trades if they offer higher market quality and competitive quotes at the time of a trade, confirming our earlier findings regarding the evolution of market share. An additional important contribution of our study is our investigation of the relationship between the pricing model chosen and the market share achieved by options exchanges during the

sample period. Specifically, we use regression specifications that control for the exchange pricing model in analyzing the relationship between market quality variables and market share. Our study advances the options market literature by demonstrating that execution cost, execution quality and quote competition continue to be determinative factors in the competition for market share over a long sample period, despite innovations in market structure, fierce fee competition, and other exchange specific factors.

The next section describes our sample. Section 3 explains the metrics used to analyze trading volume, execution costs, execution quality, and quote competition on the options exchanges. Section 4 provides a descriptive discussion of the competitive dynamics evident in the sample data. Section 5 examines the determinants of market share and Section 6 discusses the determinants of trade location for the new entrants to the options market during our sample period. Section 7 provides a summary of our findings and directions for future research.

Section snippets

Data

Our sample data was obtained from the Baruch College Options Data Warehouse, which maintains the full data feed from the Options Price Reporting Authority (OPRA). The US options exchanges included in this study are required to report to OPRA their best bid and offer prices from among market makers, firms and customer orders, trades, as well as open interest. They also are required to provide end of day summary data. We compute the National Best Bid and Offer (NBBO) by taking the highest bid and

Measures of market quality

We use market share, trade execution costs, trade execution quality, and quote competition variables across nine market centers in our study. For each variable, the monthly totals and averages over 36 months are calculated for each exchange, over all trades. Dollar volume is the sum of all trades' dollar values for a month. Volume based on number of transactions is the number of

all transactions for a month. We compute a monthly percentage market share for each exchange by dividing the dollar

Market competition

Unlike equity exchanges, which tend to focus on clienteles composed of firms within certain size and trading activity ranges, option exchanges tend to list all or most of the domestic option series for which there is market demand. Thus, competition among options exchanges tends to focus primarily on features of their market pricing models, trading platforms, and fee structures that determine trade execution cost and quality, along with the frequency and competitiveness of quotes. Five of the

Explaining market share

We further investigate competition among the exchanges using a regression framework to analyze key variables that are related to market share. We use percentage market share based on dollar trading volume as the dependent variable. Execution costs, execution quality, and quote competition are factors that can potentially explain how trading volume is shared among the exchanges.

Because exchanges use their choice of pricing model as a key factor in their competitive positioning, it is important

Determinants of trade execution location for the new entrants

The fact that our sample period contains two events that mark the introduction of new entrants to the U.S. options market provides the opportunity to analyze how the new venues establish a foothold in the market in their beginning days. Such an analysis also has the potential to confirm our findings of the evolution of market share at an aggregate level. Specifically, as a new venue competes to establish market share, it is essential to understand the factors that determine the execution of a

Summary and conclusions

In this paper we provide a review of key market microstructure performance metrics across the major options exchanges in the United States during the period between January 2008 and December 2010. We analyze the competition for trading volume, investigate the characteristics of trades and execution costs for each exchange, and examine the determinants of market share.

We find that at the beginning of 2008, ARCA and PHLX held the dominant position in the market, with around 27 percent market

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